Problem Set 3 — Block 4

Introduction to Economics · Group A6 · Universitat de Barcelona · 2026/27
Problem Set 3 · handed out Thu 5 Nov 2026 · Pavel Kocourek

This problem set is not handed in. Work through it at home and **bring your solutions to class on Thursday 12 November**: a short quiz at the start of class asks about them, and you may use them during it. The quiz is graded instead of the problem set; the sample solutions go up on the course page afterwards.

Everyone has the same problems, with slightly different numbers. Each problem comes in three variants, A, B and C: the letter next to a problem's title is your variant of that problem, and it can differ from one problem to the next. You may discuss the problems with classmates and use AI tools, but work through your own numbers yourself: the quiz asks about your copy's numbers. Keep this copy: the quiz asks for its number.

How to work. Work on your own paper and bring it to the quiz: the calculations, the tables, the graphs (draw them with a ruler; graph paper helps), and a sentence of explanation where a part asks why. Money in euros, to the cent where needed; a loss with a minus sign.

What the problems are about. Problems 1 and 2: the costs of a firm (class 11). Problem 3: a competitive market in the short and the long run (class 12). Problems 4 and 5: a monopoly and a price war (class 13, Monday 9 November).

Problem 1 — Carme's first year: accounting and economic profit

variant A

Carme makes chocolate bars in Sants. Last year, her first, she worked alone in a rented corner of a shared kitchen. She sold bars for €138 thousand and paid out €106 thousand for the rent, cocoa, sugar, wrappers and electricity. Before she started, she was a pastry chef in a hotel, earning €37 thousand a year, and she could go back to that job. She also put €50 thousand of her savings into her tools and a small tempering machine; in a bank, that money would earn 4 % a year (the tools keep their value).

(a) What was her chocolate business's accounting profit last year (in thousand euros)?

(b) What were its implicit costs, and what was its economic profit (in thousand euros a year)?

(c) Was Carme doing better or worse than in her best alternative?

Problem 2 — Carme's workshop: what its bars cost

variant B

This year Carme has moved into a workshop of her own, with one large machine that grinds and tempers the chocolate. She makes bars for shops that send their own cocoa, sugar and wrappers, so her costs are the workshop and the workers. The workshop's fixed cost (the rent, the machine, and what Carme's own time could earn elsewhere: she runs the workshop and is not one of the workers) is €270 a day, and each worker costs €120 a day. The table shows how many bars the workshop makes in a day with 0 to 6 workers.

workers	0	1	2	3	4	5	6
bars a day	0	10	60	100	125	135	140

(a) Find the marginal product of each worker. From which worker on does the marginal product fall, and why? For 1 to 6 workers, find the total cost (euros a day), the average variable cost (AVC) and the average total cost (ATC), both in euros a bar, and the marginal cost (MC) of the bars that each worker adds, in euros a bar. Round to the cent where needed. In particular: what are the AVC and the ATC with 2 workers, and the MC of the bars the 2nd worker adds?

(b) At how many bars a day is ATC lowest? (This output is the efficient scale.) What is the ATC there? Compare it with the MC of the bars the last worker adds to reach that output, and with the MC of the bars the next worker would add: why does MC cross ATC where ATC is lowest? Sketch AVC, ATC and MC, with bars a day on the horizontal axis; draw each worker's MC as a flat step over the bars that worker adds.

(c) In the long run, Carme could move to a workshop twice the size: two machines and a manager to run the second room, so twice the fixed cost, and twice as many workers as she has at the efficient scale. The bigger workshop would make 250 bars a day. What would its ATC be, compared with the ATC at the efficient scale? Does it show economies of scale, constant returns to scale or diseconomies of scale? Give one reason why a bar can cost less to make in a bigger workshop, and one reason why it can cost more.

Problem 3 — Artichokes from El Prat: a farm in a competitive market **variant B**

In winter, many small farms around El Prat sell artichokes by the box at the wholesale market. Every farm's artichokes are alike, buyers see every seller's price, and each farm sells only a tiny part of all the boxes traded. The table shows one farm's total cost in euros a day for 0 to 8 boxes a day. Like every cost in economics, it counts what the farmer gives up, her own time included. Its fixed cost (the land and the tractor, rented for the season, and her own time) is committed for the whole season, however many boxes she picks.

boxes a day	0	1	2	3	4	5	6	7	8
total cost (€ a day)	72	86	97	111	134	160	194	236	286

(a) Why does the farm take the market price as given? Today a box sells for €36. Find the marginal cost of each box. How many boxes a day should the farm sell, and what is its profit (euros a day)? Compare the price with the marginal cost of each box, and check your answer with the profit for every number of boxes.

(b) What is the farm's fixed cost? For 1 to 8 boxes, find the average variable cost (AVC) and the average total cost (ATC), in euros a box (round to the cent). Below what price would the farm stop picking for now, and below what price would it leave artichoke farming for good? Why are the two prices different?

(c) Suppose instead that the price fell to €28. How many boxes a day should the farm sell, and what is its profit (check again with the profit for every number of boxes)? Is that better than not picking at all? And if the price fell even further, to €7, how many boxes, and what profit? Would the number of boxes in (a) and (c) change if the farm's fixed cost were €20 a day higher? Why?

(d) The farm is one of 120 farms with the same costs. How many boxes a day do the 120 farms offer together at a price of €7, €28 and €36? Buyers at the market want $Q = 1440 - 20P$ boxes a day at a price of €P. Show that €36 is the market price: at €36, buyers want exactly the boxes the farms offer.

(e) At €36 the farms make a profit. In the long run, new farms with the same costs can start growing artichokes, and farms can stop; no farm's costs change as the number of farms changes. What will the price be in the long run, and how many boxes a day will each farm sell? How many boxes will buyers buy, and how many farms will there be in all? What is each farm's economic profit then, and why does no farmer leave?

Problem 4 — A patent on a tick pill

variant C

A company holds the patent on a pill that keeps ticks off dogs, so for now no one else may sell it. Making a box of pills costs it €10, however many boxes it makes, and it has no other costs. It makes the pills in batches of a thousand boxes. For each price the company could charge, the table shows how many boxes dog owners would buy in a month.

price (€ a box)	50	44	38	32	26	20	14	8
boxes a month (thousands)	1	2	3	4	5	6	7	8

(a) For each price, find the company's total revenue (thousand euros a month), and the marginal revenue of each extra thousand boxes (in euros a box). Why is the marginal revenue below the price from the second thousand on?

(b) Which price in the table should the company set (euros a box), how many boxes does it sell (thousands a month), and what is its profit (thousand euros a month)?

(c) Which quantity in the table would make the total surplus (the dog owners' gain plus the company's profit) largest, and why does the company not sell that many?

(d) Draw the demand (the table's points, or a straight line through them) and the cost of a box, and each thousand's marginal revenue as a step over that thousand; shade the deadweight loss.

Problem 5 — A price war between two ski-hire shops

variant A

Cim and Vall are the only two ski-hire shops in a Pyrenean village. Each winter month, the two shops set their prices at the same time: each sets either a high or a low price for a day's skis. Their profits, in thousand euros a month, depend on both choices (Cim's profit first, then Vall's):

		Vall's price	
		high	low
Cim's price	high	Cim 15, Vall 12	Cim 7, Vall 16
	low	Cim 18, Vall 5	Cim 10, Vall 7

(a) What is Cim's best choice if Vall sets a high price? And if Vall sets a low price? What are Vall's best choices? Does each shop have a dominant strategy?

(b) What will the two shops do, and what will each earn?

(c) Is there an outcome that is better for both? Why do they not reach it?

(d) Might they reach it if each expects to face the other month after month, with no known last month?

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